Python Logical Problem Solving Programs

1. Print Numbers from 1 to 10 Using While Loop

Question: 1. Print Numbers from 1 to 10 Using While Loop

```
num = 1

while num <= 10:
    print(num)
    num += 1
```

2. Factorial of a Number

Question: 2. Factorial of a Number

```
num = int(input("Enter a number: "))
fact = 1

for i in range(1, num + 1):
    fact *= i

print("Factorial =", fact)
```

3. Fibonacci Series Till n = 10

Question: 3. Fibonacci Series Till n = 10

```
a = 0
b = 1
n = 10

for i in range(n):
    print(a, end=" ")
    c = a + b
    a = b
    b = c
```

4. Palindrome Number

Question: 4. Palindrome Number

```
num = int(input("Enter a number: "))
original = num
reverse = 0

while num > 0:
```

```
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

if original == reverse:
    print("Palindrome")
else:
    print("Not Palindrome")
```

5. Armstrong Number

Question: 5. Armstrong Number

```
num = int(input("Enter a number: "))
original = num
total = 0

while num > 0:
    digit = num % 10
    total += digit ** 3
    num = num // 10

if original == total:
    print("Armstrong Number")
else:
    print("Not Armstrong Number")
```

6. Prime Number

Question: 6. Prime Number

```
num = int(input("Enter a number: "))

if num <= 1:
    print("Not Prime")
else:
    is_prime = True
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break

    if is_prime:
        print("Prime Number")
    else:
        print("Not Prime Number")
```

7. Palprime Number

Question: 7. Palprime Number

```
num = int(input("Enter a number: "))

original = num
reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

is_palindrome = (original == reverse)

is_prime = True
if original <= 1:
    is_prime = False
else:
    for i in range(2, original):
        if original % i == 0:
            is_prime = False
            break

if is_palindrome and is_prime:
    print("Palprime Number")
else:
    print("Not Palprime Number")
```

8. Odd or Even

Question: 8. Odd or Even

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

9. Leap Year

Question: 9. Leap Year

```
year = int(input("Enter a year: "))
```

```
if (year % 4 == 0):
   print("Leap Year")
else:
   print("Not a Leap Year")
```